Tree Reconciliation accounting for Segmental Events: A Reduced Representation of the Solution Space

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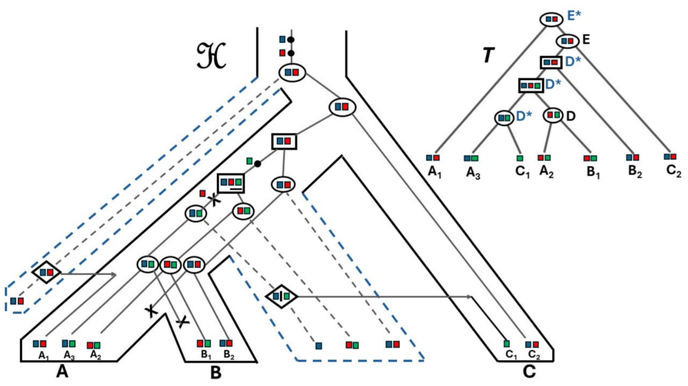
Introduction

- Synesth is a tree reconciliation algorithm explicating the evolution of genomic regions or syntenies in a set of genomes. The considered evolutionary model involves segmental events, namely events acting on multiple genes: duplications, losses, gains, synteny fissions and horizontal gene transfers which may go through unsampled or extinct species.
- Among the space of possible histories explicating a synteny and species tree given as input, we are interested in selecting the most likely ones.
- There are multiple ways to do it, for example by assigning costs (or probabilities) to event types and selecting histories minimizing (resp. maximizing) the overall cost (resp. probability). Or, by choosing the Pareto-optimal histories (in terms of counts of each event type).
- The issue is that there are still too many optimal histories. How can we find a **summarize these histories into a single "representative" one**?

Steps

➤ **Input:** A synteny tree T and a species tree S. For an illustration, we will use the example from [2] :

➤ **Synesth:** We used superrec2 (see qr-code) to calculate optimal history according to Synesth's evolutionary model [1]. The optimality criteria use here is the Pareto-optimality, in terms of the event vector of the history. Bellow is a possible history for the synteny tree and the species tree given above. The event vector for this example is $(2, 1, 0, 1, 3)$, for 2 duplications, 1 transfer duplication, 0 cut,1 transfer cut and 3 losses.



One history ($\mathcal{H}$) that extend one super reconciliation (T), of the example.

Pareto Optimal :

For any two vectors $a, b \in \mathbb{V}$:

- $a \preceq b \;\Leftrightarrow\; \forall\, i,\; a_i \leq b_i;$
- $a \prec b \;\Leftrightarrow\; \begin{cases} a \preceq b \\ a \neq b \end{cases}$

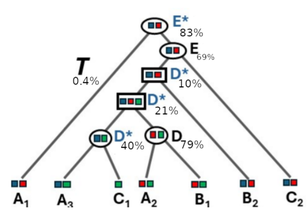
Pareto subset of $\mathbb{V}$ is

$$\{x \in \mathbb{V} \mid \neg(\exists\, y \in \mathbb{V},\; y \prec x)\}$$

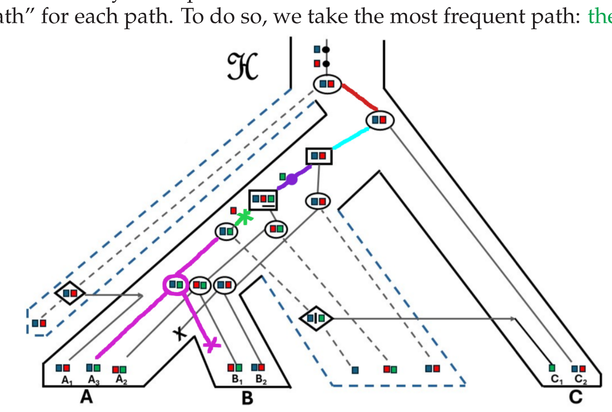
denoted $Pareto(\mathbb{V})$

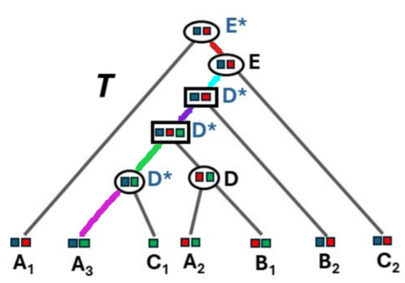
➤ **Group by Equivalence Class:** *We have too many co-optimal histories, so we want to group similar histories.*
We find multiple ways to grouping, for example by the event vector of the history. In the following, we will use a different type of group. A history extends a reconciliation (the synteny tree T in input with an event, species and synteny associated with each node). So the class is a set of "equivalent" histories, in terms of histories leading to the same event, species and synteny mapping: (e, x, s) on the nodes of the corresponding reconciliation.

➤ **Find a Representative Class:** *There are multiple classes, and now want to retain only the most "represented" ones.*
We define the probability of each class as the product of the probability of each node, which is given by the proportion of the classes with the same event at that node. We choose the classes having the highest probability.



➤ **Find a Representative History inside the Class:** *There are multiple histories in a class, so we want to summarize all the histories inside a given class as a single "representative".*
All the histories in a class have the same event, species and synteny mapping on the nodes of the synteny tree. What changes in each history is the paths between these nodes. To reconstruct a "representative history", we want to find a "representative path" for each path. To do so, we take the most frequent path: the most recurrent path in all the histories of the class.





➤ **Output:** A representative history of a representative class, among all representative histories possible in all representative classes.

Results

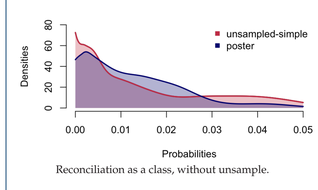
As the paths are independent one from another, changing the paths individually leads to a valid history. For the history to still belong to the same class, it also needs to be Pareto-optimal: **Is it always the case?**
We can make the following two propositions:

1. **All the representative histories for a given equivalence class are Pareto optimal.**
2. **There is at least one history, among all the representative histories for a given equivalence class, which is Pareto optimal.**

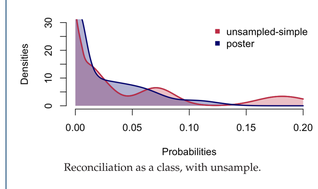
By mathematically formalizing the problem and running simulations, we could find multiple instances for which there is at least one history that is not Pareto optimal in the space of solution. Therefore, the **first proposition is false**. After several days of simulations, we couldn't find any counter-example for the second proposition. So, we tried to prove that it is true, but we didn't succeed. We therefore conjecture that the **second proposition is true**.

Numerical Simulations

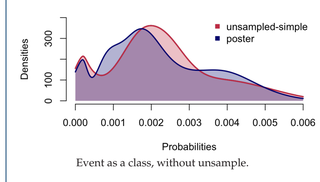
We run supperrec2 on the tree examples given in the software GitHub. minimal.in.json is the smallest example with $|S| = 2$ and $|T| = 3$, unsample-simple.in.json have $|S| = 3$ and $|T| = 7$ and poster.in.json is the example use in the poster.
Unsample is a boolean variable indicating whether or not the algorithm takes unsampled species into account. The *min*, *average* and *max* are for the probabilities (resp. the number of histories in the class) we have for the classes. Furthermore, *classes* gives us the total number of classes and *histories* the total number of histories.



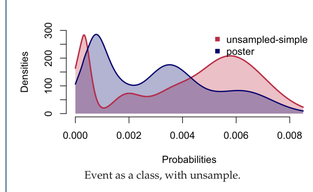
Reconciliation as a class, without unsample.



Reconciliation as a class, with unsample.



Event as a class, without unsample.



Event as a class, with unsample.

Grouping histories according to their corresponding reconciliation :

	Probability Class				
	Unsample	*Min*	*Average*	*Max*	*Classes*
minimal.in.json	false	0.333	0.556	0.667	3
unsampled-simple.in.json	false	$2.19e^{-5}$	0.016	0.047	44
poster.in.json	false	$1.53e^{-4}$	0.013	0.043	227
minimal.in.json	true	0.200	0.360	0.400	5
unsampled-simple.in.json	true	$1.3e^{-7}$	0.060	0.18	129
poster.in.json	true	$6.01e^{-6}$	0.033	0.106	229
	Number of Histories in a Class				
	Unsample	*Min*	*Average*	*Max*	*Histories*
minimal.in.json	false	1	5.7	8	17
unsampled-simple.in.json	false	1	1.2	2	51
poster.in.json	false	1	1.6	8	364
minimal.in.json	true	1	3.0	6	15
unsampled-simple.in.json	true	1	3.0	11	387
poster.in.json	true	1	3.7	16	841

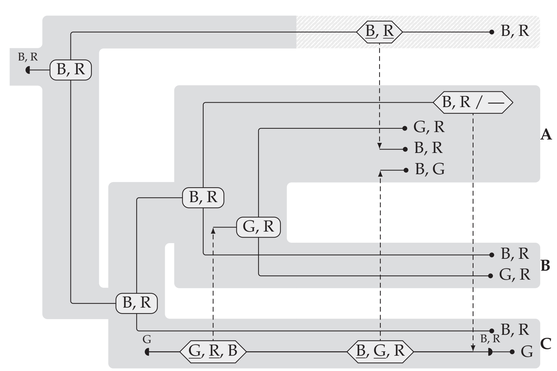
Grouping histories according to their event vector :

	Probability Class				
	Unsample	*Min*	*Average*	*Max*	*Classes*
minimal.in.json	false	$2.34e^{-2}$	0.029	0.035	4
unsampled-simple.in.json	false	$1.07e^{-4}$	0.002	0.005	26
poster.in.json	false	$1.34e^{-4}$	0.002	0.005	36
minimal.in.json	true	$1.39e^{-2}$	0.039	0.056	6
unsampled-simple.in.json	true	$1.85e^{-4}$	0.004	0.007	21
poster.in.json	true	$1.79e^{-4}$	0.003	0.007	22
	Number of Histories in a Class				
	Unsample	*Min*	*Average*	*Max*	*Histories*
minimal.in.json	false	1	4.2	8	17
unsampled-simple.in.json	false	1	2.0	4	51
poster.in.json	false	1	10.1	54	364
minimal.in.json	true	1	2.5	4	15
unsampled-simple.in.json	true	1	18.4	117	387
poster.in.json	true	2	38.2	173	841

With the clustering by reconciliation method, we find that our probabilities are more condensed on one side, and we have many classes, each containing few histories. In contrast, with the event vector clustering method, the probabilities are more spread out and we have fewer classes, each containing many more histories.

Output

For the example given in the steps section, here is the "representative history":



Future Research

➤ *Before, we conjecture that : "there is at least one history, among all the representative histories for a class, which is Pareto optimal"* **Determine whether the conjecture is true or false.**

➤ *We currently choose "arbitrarily" the representative class and history among all the representative ones.* **Find a more reliable way to select a representative class and history.**

➤ **Investigate new ways to group histories, as well as to find representative classes/histories.**

➤ **We want to find an upper bounds for the number of representative classes and histories**, to measure the effectiveness of our method for reducing the solution space.

References

superrec2

[1] Mattéo Delabre and Nadia El-Mabrouk. "Synesth: Comprehensive Syntenic Reconciliation with Unsampled Lineages". In: *Algorithms* 17 (2024), pp. 186–186. DOI: 10.3390/a17050186.
[2] Mathieu Gascon, Mattéo Delabre, and Nadia El-Mabrouk. "Simultaneously Building and Reconciling a Synteny Tree". Submitted to SPIRE. 2024.